

CASINO, Australia

WMO: 945730

Lat: 28.878S

Lon: 153.052E

Elev: 71

StdP: 14.66

Time Zone: 10.00 (AUS)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.6	41.0	27.0	21.0	62.5	30.4	24.6	58.8	19.9	67.3	17.7	65.6	4.9	280	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.8	95.4	71.6	91.7	71.7	88.5	70.5	75.9	88.4	74.6	86.0	73.5	83.6	11.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.6	121.0	80.2	71.4	116.2	79.0	70.3	111.9	78.0	39.4	88.8	38.2	86.2	37.1	83.8	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.5	14.8	33.6	102.9	2.6	4.5	31.7	106.1	30.1	108.8	28.6	111.3	26.7	114.6		
(5)			32.2	77.9	2.7	1.4	30.3	78.9	28.7	79.8	27.2	80.6	25.3	81.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.1	76.1	75.2	73.0	67.9	61.8	57.9	56.6	58.4	64.0	68.2	71.5	74.6		
	DBStd	8.14	4.48	4.06	3.61	3.42	3.74	3.89	3.96	4.23	4.80	5.09	5.31	5.04		
	HDD50	3	0	0	0	0	0	1	1	1	0	0	0	0		
	HDD65	911	0	0	0	10	110	214	262	212	75	23	4	1		
	CDD50	6232	810	707	714	538	367	238	206	260	420	564	646	762		
	CDD65	1667	345	287	249	99	13	1	2	7	45	122	200	297		
	CDH74	13233	2879	2120	1490	514	111	12	27	136	532	1186	1725	2501		
	CDH80	4668	1148	760	424	81	9	1	1	21	135	404	673	1011		
(14) Wind	WSAvg	6.4	7.1	6.9	6.1	5.5	5.2	5.8	5.8	6.3	6.6	7.2	7.4	7.2		
(15) Precipitation	PrecAvg	44.3	4.9	6.5	5.7	3.5	3.8	2.4	2.3	1.5	1.2	2.8	3.3	5.5		
	PrecMax	68.3	11.9	18.9	16.9	19.1	17.0	15.6	13.2	6.9	4.4	14.9	8.2	15.9		
	PrecMin	24.9	0.4	0.8	0.4	0.4	0.4	0.3	0.0	0.0	0.0	0.3	0.8	0.6		
	PrecStd	11.2	3.0	3.7	4.1	4.0	3.7	3.0	2.8	1.4	1.1	2.6	1.9	3.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.3	97.0	93.9	86.0	81.0	75.4	77.4	82.5	89.2	94.3	97.2	99.9		
		MCWB	74.3	74.8	72.5	68.4	64.3	61.5	59.7	60.1	63.9	66.3	68.7	71.4		
	2%	DB	93.8	91.3	87.6	82.3	76.9	72.3	73.0	77.6	83.7	88.3	91.9	93.8		
		MCWB	73.5	74.6	71.7	67.2	62.5	59.7	58.4	58.9	63.0	65.0	69.2	71.6		
	5%	DB	89.8	87.6	84.4	79.4	74.1	69.8	70.0	73.7	80.0	84.4	87.3	89.2		
		MCWB	72.6	72.4	70.7	66.4	61.5	58.5	57.0	57.4	62.0	64.7	68.7	70.8		
	10%	DB	86.1	84.5	81.6	76.7	71.5	67.2	67.2	70.4	76.4	80.6	83.2	85.5		
		MCWB	72.3	71.7	69.8	65.6	60.6	57.6	55.9	56.0	60.7	64.2	67.8	70.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.4	77.8	75.6	71.4	68.0	64.8	64.8	64.1	67.8	71.7	74.2	76.5		
		MCDB	91.3	90.4	85.7	80.9	75.5	71.3	72.7	76.1	83.4	84.7	89.4	91.0		
	2%	WB	75.8	75.9	73.7	69.6	65.5	62.6	61.8	61.6	65.6	69.1	72.0	74.7		
		MCDB	88.1	87.5	82.9	78.0	73.7	68.3	68.2	71.1	78.5	80.6	85.0	88.1		
	5%	WB	74.4	74.5	72.4	68.1	63.8	61.1	59.5	59.9	64.2	67.4	70.6	73.1		
		MCDB	85.6	84.3	80.8	76.0	70.9	66.5	66.2	68.6	75.6	78.2	82.3	84.6		
	10%	WB	73.2	73.0	71.2	66.9	62.4	59.5	57.7	58.3	62.8	65.9	69.2	71.7		
		MCDB	83.2	81.1	78.9	74.3	68.8	64.7	64.1	66.9	73.1	76.5	79.6	82.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.8	17.5	16.9	18.6	21.3	20.2	23.0	25.7	24.8	23.1	21.2	20.0		
		MCDBR	25.9	23.5	22.4	23.1	24.7	23.5	27.3	31.5	30.4	30.4	28.3	27.2		
		MCWBR	9.3	8.9	8.8	10.2	12.0	12.8	14.2	15.1	12.6	11.7	10.2	9.8		
	5% WB	MCDBR	21.3	20.3	18.3	18.9	19.7	18.1	20.3	24.5	25.3	24.0	22.8	22.5		
		MCWBR	8.3	8.4	8.0	9.3	10.4	11.0	12.3	13.3	11.6	10.5	9.1	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.389	0.374	0.356	0.343	0.313	0.307	0.299	0.312	0.354	0.381	0.386	0.396		
	taud		2.475	2.530	2.575	2.582	2.626	2.635	2.641	2.583	2.459	2.415	2.447	2.441		
	Ebn at Noon		302	301	297	286	281	276	284	292	292	296	301	301		
	Edh at Noon		37	35	31	28	24	23	23	27	34	38	38	39		
(44) All-Sky Solar Radiation	RadAvg		2049	1858	1579	1352	1118	928	1072	1359	1685	1878	2023	2044		
	RadStd		173	157	92	64	72	60	84	104	135	175	182	207		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (21 neighbors)		+1.08	N/A	N/A	+3.02	N/A	N/A	N/A	-149	+472
		+0.70	N/A	N/A	N/A	N/A	N/A	N/A	-104	+304

Nomenclature: See separate page